The empirical recurrence for OEIS A262476: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

2 September 2026

Abstract

OEIS A262476 records "Empirical recurrence of order 67", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 648$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 67, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 67 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A262476 is "Number of (6+1)X(n+1) 0..1 arrays with each row divisible by 3 and each column divisible by 5, read as a binary number with top and left being the most significant bits.". It has offset 1 and begins

$$26, 107, 2622, 22907, 552430, 8080915, 194647694, 3858564731, \ \ldots$$

The entry states:

> Empirical recurrence of order 67 (see link above)

As of the "Last modified" line on the live entry (Sep 23 2015 18:55:43, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 2187$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 648$ of the 2187, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 648$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 1296$ exact terms of the model returns order 67 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{67} c_i\, a(n-i),$$

c_1	26	c_{18}	23380353916415578776213048	c_{35}	−1599324485
c_2	3498	c_{19}	−607889201826805048181539248	c_{36}	−3475113450
c_3	−90948	c_{20}	−4603110177393848357907803460	c_{37}	90352949704
c_4	−5763570	c_{21}	119680864612240057305602889960	c_{38}	168076122893
c_5	149852820	c_{22}	766696013603583071186892274092	c_{39}	−43699791952
c_6	5960376542	c_{23}	−19934096353693159850859199126392	c_{40}	−69360339450
c_7	−154969790092	c_{24}	−108674380435571179436407853936184	c_{41}	1803368825715
c_8	−4348809157272	c_{25}	2825533891324850665346604202340784	c_{42}	2431036148787
c_9	113069038089072	c_{26}	13165230608094535499874550521914820	c_{43}	−6320693986848
c_{10}	2385545826728274	c_{27}	−342295995810457922996738313569785320	c_{44}	−7193504153804
c_{11}	−62024191494935124	c_{28}	−1367134770090021796236513016490582700	c_{45}	187031107998924
c_{12}	−1023657813764925670	c_{29}	35545504022340566702149338428755150200	c_{46}	178327417572990
c_{13}	26615103157888067420	c_{30}	121914185275408650702316590511663957380	c_{47}	−463651285689777
c_{14}	352960581951916500546	c_{31}	−3169768817160624918260231353303262891880	c_{48}	−36675327675468
c_{15}	−9176975130749829014196	c_{32}	−9342791369434969007005268393001153772995	c_{49}	9535585195621716
c_{16}	−99676915611414621281433	c_{33}	242912575605309194182136978218029998097870	c_{50}	6180025213404693
c_{17}	2591599805896780153317258	c_{34}	615124802138123972827347263318016813500730	c_{51}	−1606806555485220

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 67, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $67 + 4$ terms removed returns order 67 as well, so $\deg q_{\min} = 67 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $67 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 67 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A262476.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.